

Learning to Recover: Privileged-Actor RL for Dual-Device Neurovascular Navigation

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ABSTRACT

Reinforcement learning is increasingly demonstrated for autonomous catheter and guidewire navigation in simulation, including dual-device navigation of the cerebral vasculature. However, generalization across patient anatomy remains unresolved; agents trained on a small set of fixed vessel meshes can attain high success rates while memorizing those meshes. Training and evaluating anatomy generalization may benefit from large sets of vessels spanning a range of tortuosity, calibre, and branch geometry, which are difficult to obtain from patient segmentations alone.

Navigation is also not Markov with respect to the information available at deployment. Fluoroscopy reports device shape, but not the elastic and torsional energy stored along the shaft, the out-of-plane configuration lost to projection, or the hysteretic stick-slip contact state — quantities that govern how the device responds to the next actuation, so that visually identical configurations can evolve differently. Because these quantities are path-dependent and the projection is many-to-one, the corresponding belief is not reliably recoverable from observation history, and unobserved anatomy makes each vessel a distinct transition kernel.

The present work addresses both difficulties together — by synthesizing the anatomy distribution rather than sampling it from segmentations, and by exploiting simulator privilege during training rather than attempting to infer hidden state at inference. The finite-element solver exposes contact forces, tensions, and the full device configuration, which are Markov by construction and which mark the stress states separating the two behaviors on which navigation depends: centerline following, which governs the large majority of steps, and recovery from buckling and wrong-branch entry, which governs the few steps that determine the outcome. Geometric classifiers derived from the clinician’s planned route co-vary with these privileged fields and are available at deployment, providing deployable proxies for them. This is a property of the intended workflow rather than an assumption of privileged test-time knowledge: under supervised autonomy the route is supplied by the operator, and pre-procedural angiography and intra-procedural roadmapping already make it available. The deployable policy therefore observes only fluoroscopic device tracking and its relation to that route.

Keywords: Low-dose CT, deep learning, virtual imaging trial, lung cancer screening

1. DESCRIPTION OF PURPOSE

Endovascular thrombectomy is the standard of care for large-vessel-occlusion ischemic stroke, with pooled trial evidence demonstrating substantial functional benefit over medical management alone. Delivery depends on manually navigating a guidewire and catheter from peripheral access to the cerebral vasculature, a task that exposes the operator to cumulative ionizing radiation and that depends on a limited pool of trained neurointerventionalists, particularly outside major centres. Robotic catheter systems can execute device motions but remain teleoperated, so the navigation decisions are still made by the physician. Reinforcement learning (RL) in physics simulation has therefore been proposed for autonomous device navigation, and recent work demonstrates single- and dual-device navigation of increasingly deep cerebral targets.

The workflow comprises (1) procedural synthesis of patient-derived carotid anatomies resampled during training; (2) a geometric clearance audit admitting only anatomies whose lumen calibre matches the patient-derived vessel; (3) asymmetric actor-critic training with a privileged critic and planned-path proxies; and (4) evaluation on held-out synthetic anatomies and on the patient-derived anatomy. The simulated action space — translation and rotation of each device — is the same set of degrees of freedom a bench actuator drives, so a learned policy is executable without re-parameterization. The current study is a feasibility demonstration in simulation, intended as the foundation for staged validation: robotic actuation of those degrees of freedom on a vascular phantom, then in-vivo evaluation, before clinical translation.

2. METHODS

2.1 Procedural anatomy generation

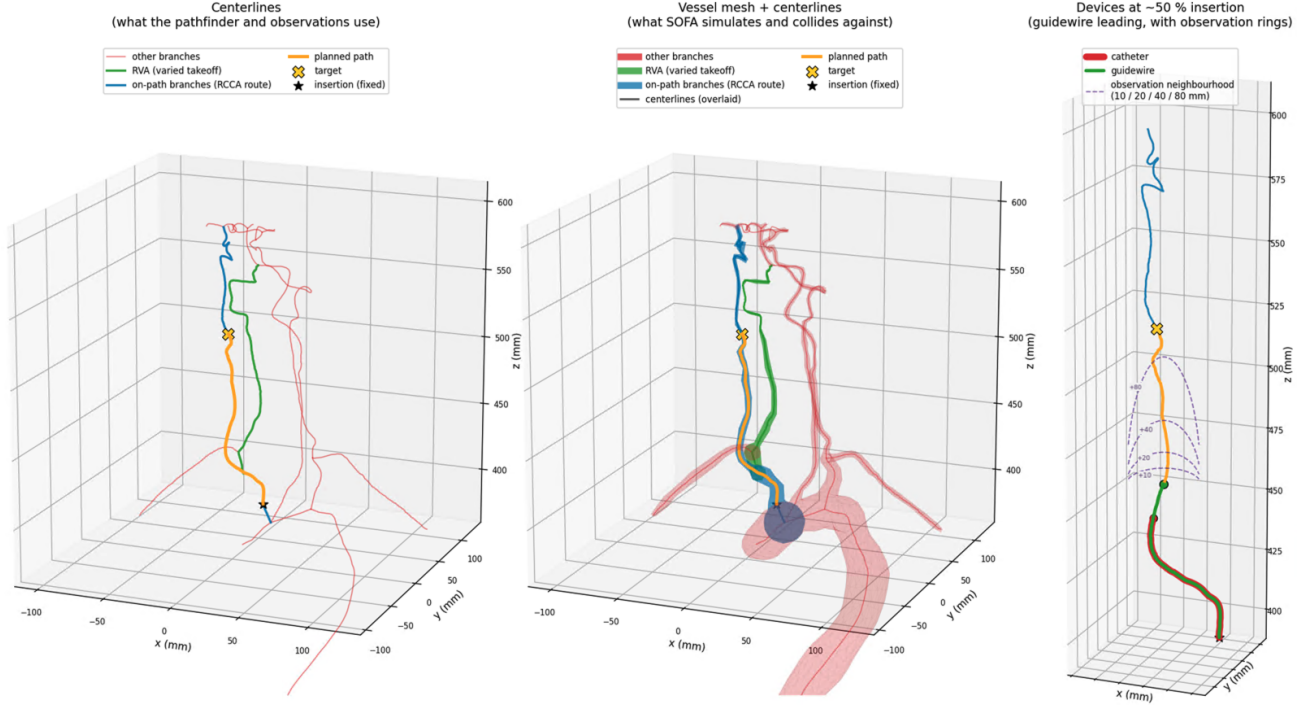


Figure 1. **Center:-** SOFA simulates Vessel mesh + centerlines. This surface is what the guidewire and catheter collide against. Vessel mesh is visible only to SOFA. The aortic arch, the descending aorta and all other branches (LCCA, LVA etc.) is fixed anatomy from a real patient. Only the RCCA and the proximal takeoff of the RVA are procedurally generated. **Left:-** Centerlines only: what the pathfinder and the observation see. The policy never observes the mesh. Everything it is given is derived from these polylines; the planned path from insertion to target, arclength progress along it, and the local geometry previews shown further in **Right:-** The catheter (red, thicker) and guidewire (green): Dashed arcs over the guidewire tip mark the neighbourhood the observation is derived from, at the actual lookahead distances.

Anatomy generation. Synthetic anatomies are produced by deforming a patient-derived tree of the carotid and vertebral branches. Only the navigated right common carotid–internal carotid axis (RCCA–RICA) and the proximal 35 mm of the adjacent right vertebral artery (RVA) are modified. The navigated centerline is displaced by three low-frequency sinusoidal modes in normalized arclength, each with random amplitude and phase, applied along two directions transverse to the vessel axis so that the course meanders smoothly. A per-anatomy tortuosity factor scales the displacement; lumen calibre is scaled independently by a few percent. The deformation tapers to zero at both ends, pinning the ostium and the terminus so that anatomically fixed landmarks are preserved and the perturbed branch remains connected to the unmodified tree. The RVA is perturbed independently, so the deflection required to select RCCA over RVA at the bifurcation differs between anatomies. Surfaces are rebuilt from the deformed centerlines each generation and audited for lumen clearance before use.

Choice of scales. The nominal displacement is 4 mm against a 2.5–3.4 mm lumen—about one vessel width of lateral wander, enough to alter the required steering while keeping the result a plausible artery. A tortuosity multiplier spanning 0.4–1.6 gives displacement scales of 1.6–6.4 mm, standing in for inter-patient variation: carotid tortuosity differs markedly between patients and is a recognized predictor of procedural difficulty. Clipping at $\pm 2\sigma$ excludes both degenerate extremes—a straight vessel that teaches nothing, and one too tortuous to be plausible or traversable. Calibre varies by only a few percent ($1\sigma \approx 7\%$), since vessel diameter varies far less between patients than vessel course does.

2.2 Simulation environment and control problem

Simulation. Finite-element simulation uses SOFA with the BeamAdapter plugin under a free-motion animation loop. **Devices.** A 0.36 mm J-tipped guidewire and a 0.6/0.7 mm microcatheter are advanced coaxially. **Action.** Four continuous commands—translation (± 30 mm) and twist (± 1.5 rad) for each device. **Observations.** The full observation is 125-dimensional and partitions into a 101-dimensional *deployable* component, which the policy receives, and a 24-dimensional *privileged* component, which is provided only to the value function during training.

The deployable component contains quantities obtainable in a catheterization laboratory: projected device tracking (40 dimensions; ten points at 15 mm spacing, tip-relative, with a two-step history), target offset in the image plane (2), the previous command (4), planned-route guidance (51), and the scripted controller’s action for the current state (4). Forty of the 51 guidance dimensions derive from the clinician’s planned route: remaining distance, cross-track offset, heading error, curvature ahead, distance and direction to the next correct daughter branch, and an on-path indicator.

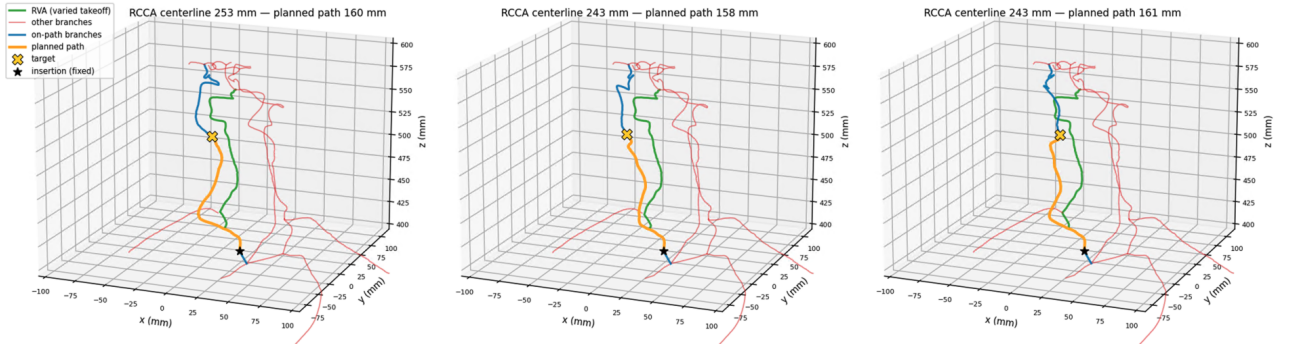


Figure 2. Representative vessel meshes with varied RCCA - RICA - siphon .

The privileged component contains simulator-only quantities: nodal velocities and forces at the distal beam elements, a contact-impulse proxy formed from the difference between constrained and free positions, accumulated axial twist, and ground-truth branch identity. These are precisely the quantities that mark stress states—buckling, wall loading, and torsional wind-up—and that are unavailable to a fluoroscopic observer.

Reward. A weighted sum of: a terminal target reward (+3.0 within 5 mm of the target); dense arclength progress along the planned route (0.01 per mm, symmetric, so that a round trip nets zero) with a lateral-deviation penalty (0.001 per mm); a small per-step cost varying by path segment (−0.007 to 0); discrete ± 1.0 events on committing to a correct or incorrect daughter branch at a fork; and a potential-based anti-buckling term (coefficient 0.5) formed from guidewire slack and mean contact displacement. Episodes terminate on target contact and truncate at 600 control steps or on simulator error; vessel-end truncation is disabled so that off-path excursions remain recoverable rather than fatal.

2.3 Asymmetric actor-critic

Because the deployable observation is not Markov (Sec. 1), a value function estimated from it alone must average over the hidden mechanical state, which both biases the estimate and inflates its variance. We therefore train an asymmetric actor-critic: the critics receive the full 125-dimensional observation, while the policy receives only the 101-dimensional deployable component. Both are optimized jointly in a single run, with no separate imitation stage.

Formally, the critics estimate $Q_{\theta}(o^{\text{priv}}, a)$ while the policy is $\pi_{\phi}(a \mid o^{\text{dep}})$, where o^{dep} is the prefix of o^{priv} . The policy gradient is taken through the critic evaluated at the privileged observation, but the action is produced from the deployable observation alone. The critic thus supplies a better-conditioned learning signal—its regression target is a function of quantities that are Markov by construction—without ever appearing at inference. At deployment the critics are discarded and only π_{ϕ} is executed, so the policy is deployable by construction and there is no distillation gap between the trained and the deployed controller.

Algorithm. Training uses soft actor-critic with two critics and Polyak targets ($\tau = 0.005$, $\gamma = 0.99$), taking $\min(Q_1, Q_2)$ in both the target and the actor loss. The entropy term is omitted from the critic backup while retained in the actor objective. The critics are two-layer multilayer perceptrons of width 256 taking the privileged observation and the action; the policy is a two-layer network of the same width taking the deployable observation. Layer normalization is applied in the critic bodies only. Optimization uses Adam at 3×10^{-4} with gradient-norm clipping at 1.0. The entropy temperature is auto-tuned against a target entropy of +1.0, rather than the conventional $-n_{\text{actions}}$, with $\log \alpha$ railed to $[-5, 0]$; the policy log-standard-deviation is soft-bounded to $\sigma \in (0.135, 1.0)$.

2.4 Correspondence between planned-path geometry and privileged state

An asymmetric critic improves the policy only if the deployable observation is rich enough to express the behavior that the privileged value estimates favor. If the mechanical states that determine value were invisible to the policy, a better-conditioned critic would grade actions the policy cannot condition on, and the asymmetry would yield no benefit. We therefore test this correspondence directly rather than assume it.

A probe is trained to predict privileged stress quantities—contact-impulse magnitude and distal nodal force—from the deployable planned-path features alone, on states sampled from policy rollouts across audited anatomies. Performance is reported as area under the receiver operating characteristic curve for a binarized high-contact state, and as coefficient of determination for the continuous targets, each against a shuffled-label control.

Two constraints make the test meaningful. First, any deployable feature that is numerically identical to a privileged dimension is excluded from the probe: guidewire slack, in particular, appears on both sides of the split and is computed from the same path projection, so including it would measure an identity rather than a correspondence. Second, the probe is fitted and evaluated on audited anatomies only, since geometry the device cannot traverse produces contact signatures that no deployable feature should be expected to explain.

The result quantifies the extent to which planned-path geometry acts as an observable proxy for unobservable mechanical state, and therefore predicts how much of the privileged critic’s advantage the deployable policy can realize.

2.5 Behavioral analysis

Navigation performance is decomposed into two behaviors by post-hoc analysis of the complete step logs. A *stall* opens when the guidewire is advanced above 2 mm/s while path progress remains flat within 0.3 mm for twelve consecutive steps, using a decaying counter so that isolated command reversals do not close it. A stall resolves when progress passes the frontier at which it began, and is classified by the guidewire length actually withdrawn—under 1 mm, 1–8 mm, or over 8 mm—measured from executed insertion rather than commanded retraction, since a retraction command against a wedged device moves nothing.

Three success measures are separated because they occupy different units of analysis: episode success (per episode), stall-resolution rate (per stall event), and the conversion of resolved stalls into successful episodes (per episode). Absolute rates move by roughly 20 percentage points across reasonable threshold choices, so only differences and trends are reported, and the analysis is restricted to audited anatomies.

2.6 Evaluation protocol

Evaluation uses 98 fixed seeds across 50 procedurally generated anatomies, redrawn every two episodes across 16 workers, with the clearance audit of Sec. 2.1 applied at generation time. Actions are taken deterministically as the hyperbolic tangent of the policy mean, from the deployable observation only. An episode succeeds if the guidewire tip comes within 5 mm (three-dimensional Euclidean distance) of a target sampled at least 40 mm into the RCCA, within 600 control steps.

Anatomy identity is established by hashing branch-coordinate arrays. This is necessary because the obvious identifiers are unreliable: the planned path moves with the target, and the mesh fingerprint is built from a seed that is reassigned on reset, so either can report a match between geometrically different vessels.

Results are stratified by planned-path length into three bands, with cut points at 146 mm and 210 mm, corresponding approximately to the common carotid artery, the mid-internal carotid artery, and the cavernous

segment. These are cut points on path length rather than anatomical segmentation; the modelled vessel terminates at the internal carotid artery terminus, and the middle cerebral artery is not represented.

Transfer is assessed on the patient-derived anatomy using its original segmented surface rather than a reconstruction, with all other settings unchanged.

Binomial (Wilson) confidence intervals are reported. These understate the true uncertainty: episodes are not independent, as 16 persistent simulator processes each run six to seven sequential episodes, and the initial device twist is unseeded, so repeat launches are not bit-reproducible.

3. RESULTS

Table 1. Evolution of navigation success and stall handling over training, for both configurations. Bins are equal-episode-count quantiles within each run (15k episodes over 4m steps). Episode success is measured per episode; stall resolution is measured per stall event. In Configuration A the stall-resolution rate falls by 24 percentage points while episode success rises, indicating that the improvement is obtained by avoiding stalls rather than by resolving them.

Measure	Config.	Training-progress bin								Δ
		1	2	3	4	5	6	7	8	
Episode success (%)	A	52	56	60	58	61	60	62	59	+7
	B	46	54	40	55	53	57	61	56	+10
Stall resolution (%)	A	65	58	53	58	46	52	39	41	−24
	B	68	44	72	73	75	64	66	63	−5

Table 1 shows that the configurations reach comparable navigation success during training phase by 2 means; one with asymmetric actor critic (B) and one without (A). Table 2 reports navigation success on 50 held-out synthetic anatomies, and on the patient-derived anatomy, both stratified by target depth. The per-event column separates the two training runs. In the baseline, the fraction of stalls the controller escapes falls from 65% to 41%—a 24-point decline—over exactly the interval in which episode success rises. The proposed method sustains stall resolution between 63% and 75% throughout, declining by 5 points. The baseline therefore improves by *not entering* stalls rather than by escaping them, while the proposed method improves without surrendering the ability to escape.

Read together, Tables 1–2 indicate that the asymmetric actor critic measures move together with transfer. Within Configuration B, the later checkpoint improves on both measures simultaneously namely episode success and stall resolution and improves on both validation set comprising of held-out synthetic anatomy and on the test set of the real patient anatomy. Improvements confined to centerline following, without a corresponding gain in stall resolution, produced the smaller transfer gain.

Table 2. Navigation success on 50 held-out synthetic anatomies (98 episodes, deterministic policy, 600-step limit, 5 mm success threshold), stratified by target depth along the planned path. Transfer to the patient-derived anatomy, evaluated on its original segmented surface with all other settings unchanged. No model was exposed to this anatomy during training.

Model	Validation set			Test set		
	50 held-out synthetic anatomies			Patient-derived anatomy		
	Overall	Mid-ICA	Cavernous	Overall	Mid-ICA	Cavernous
Configuration A	68.4%	72.6%	12.1%	52.0%	58.5%	6.7%
Configuration B	84.7%	100%	53.1%	75.5%	90.2%	53.3%

4. CONCLUSIONS

We conclude three observations based on our experiment. First, training-phase episode success is not a sufficient statistic for the quality of a improving policy. Second, the loss of recovery is a property of the training dynamics, not of capacity or of the anatomy distribution. Third, sustained stall resolution, and not training-phase episode success, is what tracks transfer to unseen anatomy. The tables also serve as the ablation of the privileged component. Removing it returns the plain baseline, and the effect of that removal is not a uniform degradation but a specific one: episode success is essentially preserved, while stall resolution collapses.

REFERENCES